RUSEKERE SECONDARY SCHOOL
S.5 MATHEMATICS PAPER ONE END OFTERM III 2025
TIME: 3 HOURS

SECTION A (Answer all items in this section)

Item 1

A road construction company is designing a curved embarkment whose cross section follows the equation $\boldsymbol{4x^2 + 3y^2 = 12}$. To determine the maximum safe slope of a supporting beam, an engineer draws a straight line representing the beam in the form $\boldsymbol{y = mx + c}$. for the safety, the beam must just touch the embarkment exactly one point, without cutting through it.

Task

Given the supporting beam must be tangent to the embarkment curve, show that the relationship between the constant c and the gradient m is; $\boldsymbol{4 + 3m^2 = c^2}$.

Item 2

A radio engineer is analyzing the combined signal produced by two wave generators. One generator produces a signal proportional to $sin\theta$, while the other produces a signal proportional to $cos\theta$. Due to equipment setting, the resulting output voltage at time θ seconds is modeled by; $V = 2\sin\theta + 3\cos\theta$. To compare this signal with a standard waveform used in testing, the engineer wants to re rewrite the expression in the form; $V = Rsin(\theta + \alpha)$, where R is the amplitude of the combined signal and α is the phase shift.

Task

Express $2\sin\theta + 3\cos\theta$ in the form $Rsin(\theta + \alpha)$.

Item 3

A company manufactures metal rods whose flexibility depends on the temperature factor x. The safety of the rod is guaranteed only if the following stability index S satisfies; $S = \frac{7-2x}{(x+1)(x-2)} > 0$. Engineers must determine the range of temperatures x for which the rods remain safe during production.

Task

Help the engineer to solve the stability index S.

Item 4

A scientist is muddling the vibration of a mechanical system whose displacement at time x seconds is given by; $y = e^{x} cos3x$. During testing, the system is expected to obey the differential equation, $\frac{d^2y}{dx^2} - 2\frac{d^2y}{dx^2} + 10y = 0$. To satiety that the theoretical model matches the actual behavior of the system,

Task

Help the scientist check if the displacement function satisfies the equation as shown above.

Item 5

A quality- control technician in a laboratory is testing the purity of a newly synthesized chemical compound. The purity level of the compound is represented by a mathematical expression that helps compare it with the standard compound. After performing several reactions, the purity indicator for the sample is found to be, $P = 23 - 4\sqrt{15}$. To classify the compound, the technician must compute the square root of this value, because the purity scale used in the lab is based on the square root units.

Task.

Determine the square root of the expression above.

Item 6

An engineer is designing a vibration system for a machine. The motion of the machine is designed using quadratic equations;
$ax^2 + bx + c = 0$ and $x^2 + x + d = 0$. For the system to operate safely, the two vibrations must reach their maximum displacement at exactly the same value of x. This means the quadratic equations must have a common root. The engineer wants to verify a mathematical condition that guarantees this shared displacement.

Task

Show that if the two equations have a common root, then relationship below must hold. $(c - ad)^2 = (b - a)(c - bd)$.

SECTION B (Answer 3 Items)

Item 7

A chemical engineer is studying the rate at which a certain substance flows through a reaction chamber. The flow rate at time x seconds is muddled by the function; $f(x) = \frac{3x^2+2x^2-3x+1}{x(1-x)}$ to calculate the total amount of substance that passes through the chamber over a short time interval, the engineer needs to integrate this rate function. However, the function is complicated to integrate directly and must first be written using partial fractions.

Task

a) Help the engineer in expressing the fraction in partial fractions
b) Help the engineer in integrating the fraction.

Item 8

A civil engineer is designing a suspension footbridge whose supporting cable follow a mathematical curve for aesthetic and structural reasons. The vertical displacement y of the cable (in meters) from its central horizontal support is modeled by the function, $y = \frac{1}{4x^2+1}$, where x is the horizontal distance from the center of the bridge. The engineer needs to fully understand the behavior of this curve to determine whether it is suitable for construction.

a) Help the engineer to determine the coordinates of the turning points of the cables curve.
b) Determine the equations of all the asymptotes of the curve and explain their meaning in the context of the bridge.
c) Using the results above, sketch the curve and comment on whether this shape is practical fir suspension bridge.

Item 9

A telecommunication company is analyzing a signal that varies according to the trigonometric relationships. When the signal passes through a certain electronic filter, its phase angle Θ is transformed in a way that must be expressed mathematically. The engineer discovers that the resulting signal amplitude is proportional to an expression involving $tan3\theta$. To validate the design of the filter, she must show that this transformation follows a known trigonometric identity.

a) Show that the following identity holds for the transformed phase angel, $tan3\theta = \frac{tan\theta(3-tan^2\theta)}{1-3tan^2\theta}$
b) Later, the engineer notices that the combined effect of several oscillations in the signal produces an equation,

$cos4x + cos6x + cos2x = 0$, for the values $0° \leq x \leq 180°$ help her to determin all the possible values of x that cause the signal to cancel out.

Item 10

A physics student is studying how electrical resistance changes along a specially designed nano-wire. The resistance at a point x along the wire is modeled by the function that depends on the ratio of the square of the position to the quadratic expression. To determine the total electrical effect across a segment of the wire, the student needs to evaluate two important integrals.

a) The instantaneous resistance density at position x is given by; $R(x) = \frac{x^2}{x^4-1}$. To find the net resistance contribution, the student must compute; $\int \frac{x^2}{x^4-1} dx$

b) Later, the student analyses how resistance behaves as current follows from $x = 0$ and $x = 1$. The relative function is ;$I(x) = \frac{x}{\sqrt{(1+x)}}$. Determine the total current affected by evaluating;

$$\int_0^1 \frac{x}{\sqrt{(1+x)}} dx$$

END